

Synthetic paper: Two logit paths

Physical page 1; printed page i.

$$y = \text{softmax}(a + b) V.$$

Both paths contribute to the same normalized weights.

Example and a claim

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$a=(0,0)$; $b=(0,0)$; $V=(0,10)$. Output is 5.

Change b to $(0, \log(3))$. Output is 7.5.

Claim: this proves RoPE always amplifies error in every real model.

Appendix: scope

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This is an invented teaching example, not empirical evidence.

No RoPE-specific model, data or experiment was studied.